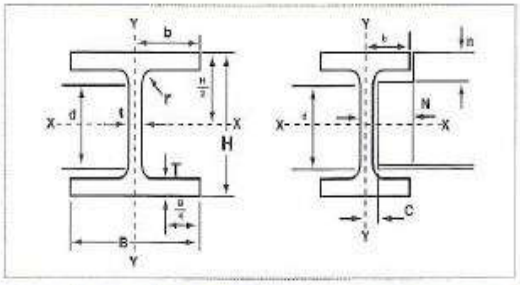
Dimensions and Properties

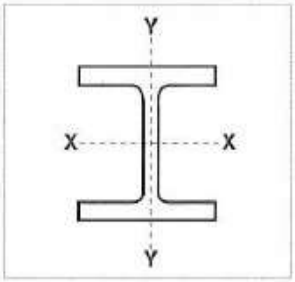


4a (i) Beams (Imperial Units) - Part I

	Designation Size		Unit Weight		Height Of Beam	Width Of Beam	Thickness		Root Radius	Depth Between Fillets	Area Of Section	Ratios For Local Buckling		Dimensions For Detailing		
							Flange	Web				Flange	Web	End Clearance	Notch	
					H	B	T	t	r	d	A	b/T	d/t	C	N	n
	in	(mm)	lb/ft	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
W12	12 x 6 1/2	(305 x 165)	21	31.3	306.0	164.0	7.4	5.0	7.6	275.2	39.4	11.08	55.0	5	90	15
			26	38.7	310.4	164.8	9.7	5.8	7.6	275.9	49.3	8.54	47.2	5	89	17
			27	40.2	303.4	165.0	10.2	6.0	8.9	265.2	51.3	8.09	44.2	5	90	19
			30	44.6	313.4	165.6	11.1	6.6	7.6	276.0	56.5	7.46	41.8	5	90	19
			31	46.1	306.6	165.7	11.8	6.7	8.9	265.2	58.7	7.02	39.6	5	90	21
			35	52.1	317.5	166.6	13.2	7.6	7.6	275.9	66.7	6.31	36.2	6	89	21
			36	53.6	310.4	166.9	13.7	7.9	8.9	265.2	68.8	6.09	33.6	6	90	23
	12 x 8	(305 x 203)	40	59.5	303.3	203.3	13.0	7.5	15.2	246.9	75.6	7.82	33.0	6	108	28
			45	67.0	306.3	204.3	14.6	8.5	15.2	246.7	85.2	7.00	29.0	6	108	30
			50	74.4	309.6	205.2	16.2	9.4	15.2	246.8	94.5	6.33	26.3	7	108	31
	12 x 10	(305 x 254)	53	78.9	306.3	253.9	14.6	8.8	15.2	246.7	100	8.70	28.2	6	133	30
			58	86.3	309.6	254.3	16.2	9.1	15.2	246.8	110	7.85	27.0	7	133	31
W14	14 x 5	(356 x 127)	22	32.7	349.0	125.4	8.5	6.0	10.2	311.6	42.1	7.38	51.9	5	70	19
			26	38.7	353.4	126.0	10.7	6.6	10.2	311.6	49.8	5.89	47.2	5	70	21
	14 x 6 3/4	(356 x 171)	30	44.6	351.4	171.1	9.7	7.0	10.2	311.6	57.3	8.82	44.5	6	92	20
			34	50.6	355.0	171.5	11.5	7.4	10.2	311.6	64.9	7.46	42.1	6	92	22
			38	56.6	358.0	172.2	13.0	8.1	10.2	311.6	72.6	6.62	38.5	6	92	23
			45	67.0	363.4	173.2	15.7	9.1	10.2	311.6	85.5	5.52	34.2	7	92	26
	14 x 8	(356 x 203)	43	64.0	347.0	203.1	13.4	7.8	15.2	289.8	81.2	7.58	37.4	6	108	29
			48	71.4	350.3	204.0	15.1	8.6	15.2	289.7	91.2	6.75	33.5	6	108	30
			53	78.9	353.6	204.7	16.7	9.4	15.2	289.8	100	6.13	30.8	7	108	32
	14 x 10	(356 x 254)	61	90.8	352.8	253.9	16.3	9.5	15.2	289.8	115	7.79	30.4	7	132	32
			68	101.2	356.6	254.8	18.3	10.5	15.2	289.6	129	6.97	27.5	7	132	33
			74	110.1	359.9	255.8	19.9	11.4	15.2	289.6	141	6.41	25.3	8	132	35
			82	122.0	363.5	257.3	21.7	12.9	15.2	289.7	155	5.92	22.5	8	132	37
	14 x 14 1/2	(356 x 368)	87	129.5	355.6	368.6	17.5	10.4	15.2	290.2	164	10.5	27.9	7	189	33
			90	133.9	356.1	368.8	18.0	11.2	15.2	289.6	171	10.2	25.9	8	189	33
			99	147.3	359.7	370.0	19.8	12.3	15.2	289.7	188	9.34	23.5	8	189	35
			103	153.3	362.0	370.5	20.7	12.3	15.2	290.2	195	8.95	23.6	8	189	36
			109	162.2	363.7	371.0	21.8	13.3	15.2	289.6	207	8.49	21.7	9	189	37
			119	177.1	368.2	372.6	23.8	14.4	15.2	290.2	226	7.83	20.2	9	189	39
			120	178.6	367.8	372.6	23.9	15.0	15.2	289.6	228	7.80	19.3	9	189	39
			132	196.4	372.4	374.0	26.2	16.4	15.2	289.7	250	7.15	17.7	10	189	41
			136	202.4	374.6	374.7	27.0	16.5	15.2	290.2	257	6.94	17.6	10	189	42
	14 x 16	(356 x 406)	118	175.6	360.0	396.0	20.0	20.0	15.0	290.0	224	9.90	14.5	12	198	35
			145	215.8	375.4	393.7	27.7	17.3	15.2	289.6	275	7.11	16.8	11	198	43
			158	235.1	381.0	394.8	30.2	18.4	15.2	290.2	299	6.54	15.8	11	198	45
			159	236.6	380.5	395.4	30.2	18.9	15.2	289.6	302	6.54	15.3	11	198	45
			176	261.9	386.6	397.5	33.3	21.1	15.2	289.7	334	5.97	13.7	13	198	48
			193	287.2	393.6	399.0	36.5	22.6	15.2	290.2	366	5.47	12.8	13	198	52
			211	314.0	399.3	401.3	39.6	24.9	15.2	289.7	400	5.06	11.6	14	198	55
			219	325.9	403.1	402.0	41.2	25.5	15.2	290.3	415	4.88	11.4	15	198	56
			228	339.3	406.4	403.0	42.9	26.6	15.2	290.2	433	4.70	10.9	15	198	58
			233	346.7	407.4	403.6	43.7	27.2	15.2	289.6	442	4.62	10.7	16	198	59
			257	382.5	416.1	406.3	48.0	29.8	15.2	289.7	488	4.23	9.71	17	198	63
			264	392.9	419.0	407.0	49.2	30.6	15.2	290.2	501	4.14	9.48	17	198	64
			283	421.1	425.2	409.2	52.6	32.8	15.2	289.6	537	3.89	8.84	18	198	68
			311	462.8	434.8	412.2	57.4	35.8	15.2	289.6	590	3.59	8.09	20	198	73
			314	467.3	436.6	412.2	58.0	35.8	15.2	290.2	595	3.55	8.11	20	198	73

Note: Calculated based on 1 kg = 2.2046 lb and 1 m = 3.2808 feet.
*Please contact us for further clarification.

Dimensions and Properties



4a (i) Beams (Imperial Units) - Part II

Unit Weight		Surface Area per Metre	Second Moment Of Area		Radius Of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant	Designation Size	
			Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y						
lb/ft	kg/m	m^2	cm^4	cm^4	cm	cm	cm^3	cm^3	cm^3	cm^3	u	x	H	J	In	
21	31.25	1.24	6554	544	12.9	3.72	428	66.4	476	102	0.881	42.4	0.121	6.3	12 x 6 1/2	W12
26	38.69	1.26	8498	720	13.1	3.82	548	87.4	609	134	0.887	33.8	0.163	12.5		
27	40.18	1.24	8503	764	12.9	3.86	560	92.7	623	142	0.889	31.0	0.164	14.7		
30	44.64	1.26	9864	841	13.2	3.86	629	102	703	156	0.887	29.7	0.192	18.7		
31	46.13	1.25	9899	896	13.0	3.90	646	108	720	166	0.891	27.1	0.195	22.2		
35	52.09	1.27	11860	1019	13.3	3.91	747	122	838	188	0.889	25.3	0.236	30.8		
36	53.57	1.26	11700	1063	13.0	3.93	754	127	846	196	0.889	23.6	0.234	34.8		
40	59.53	1.38	12840	1823	13.0	4.91	847	179	938	274	0.890	22.9	0.384	39.1	12 x 8	
45	66.97	1.39	14570	2078	13.1	4.94	952	203	1060	311	0.889	20.6	0.442	54.6		
50	74.41	1.40	16350	2337	13.2	4.97	1056	228	1183	349	0.889	18.8	0.503	73.5		
53	78.87	1.58	17700	3986	13.3	6.30	1156	314	1276	477	0.876	20.4	0.848	65.6	12 x 10	
58	86.31	1.59	19740	4444	13.4	6.36	1275	349	1411	531	0.878	18.7	0.956	86.5		
22	32.74	1.17	8250	280	14.0	2.58	473	44.7	543	70.3	0.863	42.2	0.0813	8.79	14 x 5	W14
26	38.69	1.18	10170	358	14.3	2.68	576	56.8	659	89.1	0.871	35.2	0.105	15.1		
30	44.64	1.36	12070	811	14.5	3.76	687	94.8	775	147	0.874	36.8	0.237	15.8	14 x 6 3/4	
34	50.60	1.36	14140	968	14.8	3.86	796	113	896	174	0.881	32.1	0.286	23.8		
38	56.55	1.37	16040	1108	14.9	3.91	896	129	1010	199	0.882	28.8	0.330	33.4		
45	66.97	1.38	19460	1362	15.1	3.99	1071	157	1211	243	0.886	24.4	0.412	55.7		
43	63.99	1.46	17760	1874	14.8	4.80	1024	185	1138	283	0.891	25.9	0.521	43.2	14 x 8	
48	71.43	1.47	20170	2140	14.9	4.84	1151	210	1285	322	0.891	23.3	0.601	60.4		
53	78.87	1.48	22480	2391	15.0	4.88	1271	234	1424	359	0.892	21.3	0.679	80.1		
61	90.78	1.68	26540	4451	15.2	6.21	1505	351	1668	534	0.886	21.5	1.26	90.3	14 x 10	
68	101.19	1.69	30060	5048	15.3	6.26	1686	396	1878	604	0.886	19.4	1.44	126		
74	110.12	1.69	33120	5569	15.3	6.29	1840	435	2058	665	0.886	18.0	1.61	161		
82	122.03	1.70	36680	6175	15.4	6.31	2018	480	2270	734	0.884	16.6	1.80	211		
87	129.47	2.14	40250	14610	15.6	9.43	2264	793	2479	1199	0.844	19.9	4.18	153	14 x 14 1/2	
90	133.93	2.14	41580	15080	15.6	9.40	2335	818	2565	1238	0.842	19.2	4.31	169		
99	147.33	2.15	46240	16730	15.7	9.43	2571	904	2838	1370	0.842	17.7	4.83	223		
103	153.28	2.16	48590	17550	15.8	9.49	2685	948	2965	1435	0.844	17.0	5.11	251		
109	162.21	2.16	51540	18600	15.8	9.49	2834	1003	3142	1519	0.843	16.2	5.43	296		
119	177.09	2.17	57120	20530	15.9	9.54	3103	1102	3455	1671	0.844	15.0	6.09	381		
120	178.58	2.17	57290	20600	15.9	9.51	3115	1106	3475	1678	0.842	14.9	6.09	390		
132	196.44	2.18	63720	22820	16.0	9.55	3422	1221	3838	1853	0.843	13.7	6.84	511		
136	202.39	2.19	66260	23690	16.1	9.60	3538	1264	3972	1920	0.844	13.4	7.16	558		
118	175.60	2.24	51770	20730	15.2	9.61	2876	1047	3235	1603	0.810	16.1	5.99	322	14 x 16	
145	215.78	2.26	71240	28180	16.1	10.1	3796	1432	4264	2172	0.832	13.0	8.52	632		
158	235.13	2.28	79090	30990	16.3	10.2	4151	1570	4687	2383	0.834	12.1	9.54	812		
159	236.62	2.28	79160	31170	16.2	10.2	4161	1577	4702	2394	0.833	12.0	9.56	822		
176	261.92	2.29	89040	34860	16.3	10.2	4606	1754	5244	2667	0.833	11.0	10.9	1101		
193	287.22	2.31	99880	38680	16.5	10.3	5075	1939	5812	2949	0.835	10.2	12.3	1441		
211	314.00	2.33	110500	42720	16.6	10.3	5537	2129	6387	3243	0.834	9.5	13.8	1854		
219	325.91	2.34	116400	44660	16.7	10.4	5777	2222	6681	3384	0.836	9.2	14.6	2073		
228	339.30	2.35	122500	46850	16.8	10.4	6031	2325	6999	3544	0.836	8.8	15.5	2343		
233	346.74	2.35	125100	47930	16.8	10.4	6141	2375	7140	3621	0.836	8.7	15.9	2478		
257	382.46	2.37	141500	53750	17.0	10.5	6803	2646	7975	4038	0.837	8.0	18.2	3292		
264	392.87	2.38	146600	55370	17.1	10.5	6998	2721	8222	4154	0.837	7.9	18.9	3545		
283	421.15	2.40	159800	60150	17.2	10.6	7516	2940	8887	4492	0.837	7.4	20.9	4336		
311	462.82	2.42	180100	67130	17.5	10.7	8283	3257	9877	4983	0.838	6.9	23.9	5655		
314	467.28	2.42	183000	67830	17.5	10.7	8383	3291	10000	5034	0.839	6.9	24.3	5809		

Note: Calculated based on 1 kg = 2.2046 lb and 1 m = 3.2808 feet.
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